INDUSTRIAL DIESEL GENERATOR MONITORING AND PREVENTIVE MAINTENANCE SYSTEM

ASSIGNMENT – 12

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PROJECT OVERVIEW

The Industrial Diesel Generator Monitoring and Maintenance System is designed to provide continuous monitoring of an industrial diesel generator and improve its reliability during operation. Diesel generators are widely used as backup and continuous power sources in industries, hospitals, commercial buildings, and other critical facilities. Unexpected generator failures can result in power interruptions, production losses, and high maintenance costs. Therefore, continuous condition monitoring is important for maintaining reliable operation.

The proposed system uses sensors to collect important operating parameters such as engine temperature, oil pressure, fuel level, vibration, RPM, generator voltage, and current. These parameters provide useful information about the health and performance of the generator. An ESP32 microcontroller receives the sensor readings and processes them to determine whether the generator is operating within the safe operating range.

The monitored values are displayed on an LCD/OLED display so that the operator can easily observe the generator condition. If any parameter reaches an abnormal level, the system provides an immediate warning through an LED and buzzer. This helps the operator identify potential problems before they develop into major failures.

The system can also use Wi-Fi connectivity to transmit generator data to an IoT cloud platform. The cloud dashboard allows users to remotely monitor generator parameters, observe trends, store historical data, and identify changes in performance. This makes the system useful for industrial applications where the generator may be located away from the maintenance or control room.

The collected data can support preventive maintenance by helping maintenance personnel determine when inspection or servicing is required. Instead of waiting for a component to fail, maintenance activities can be planned based on the monitored condition of the generator.

Key Points

- Continuous monitoring of diesel generator operating parameters.
- Measurement of temperature, oil pressure, fuel level, vibration, RPM, voltage, and current.
- ESP32-based data acquisition and processing.
- Real-time parameter display using LCD/OLED.
- Automatic fault and abnormal-condition detection.
- LED and buzzer alerts for critical conditions.
- Wi-Fi-based remote monitoring through an IoT cloud platform.
- Storage and analysis of historical generator data.
- Supports preventive and condition-based maintenance.
- Helps reduce unexpected breakdowns and maintenance downtime.
- Improves generator safety, reliability, and efficiency.
- Helps increase the service life of generator components.

PROBLEM STATEMENT

Industrial diesel generators are essential for providing continuous and backup power in industries and critical facilities. However, conventional generators are often monitored manually, making it difficult to identify developing faults at an early stage. Problems such as overheating, low oil pressure, excessive vibration, low fuel level, abnormal RPM, voltage fluctuations, and overload can lead to serious equipment damage and unexpected power failure.

The absence of continuous monitoring and timely maintenance can increase operating costs, maintenance expenses, and equipment downtime. Therefore, there is a need for an automated monitoring system that can continuously observe the generator's operating parameters and provide immediate alerts when abnormal conditions are detected.

The proposed system uses sensors and an ESP32 microcontroller to collect and process generator parameters in real time. The system displays the operating status locally and can transmit the data to a cloud platform for remote monitoring and analysis. It also supports preventive maintenance by identifying abnormal trends and providing early warnings.

Key Problem Areas

- Manual monitoring of generator parameters is time-consuming.
- Faults may not be detected at an early stage.
- Overheating can cause engine damage.
- Low oil pressure can lead to poor lubrication and component failure.

- Excessive vibration may indicate mechanical problems.
- Low fuel levels can interrupt generator operation.
- Abnormal voltage, current, or RPM can affect generator performance.
- Unexpected failures can cause costly downtime.
- Lack of historical data makes performance analysis difficult.
- Preventive maintenance is often performed based on fixed schedules rather than actual operating conditions.
- A real-time IoT-based monitoring system is required to improve reliability and maintenance planning.

PROBLEM SOLUTION

The proposed Industrial Diesel Generator Monitoring and Maintenance System provides an automated solution for continuously monitoring generator performance and detecting abnormal operating conditions. The system uses sensors connected to an ESP32 microcontroller to collect real-time information from the generator.

The sensor data is processed and compared with predefined safe operating limits. When a parameter exceeds its limit, the system immediately generates an alert through a buzzer and LED, allowing the operator to take corrective action before serious damage occurs.

The system also supports IoT-based remote monitoring by transmitting the collected data to a cloud platform through Wi-Fi. Operators can view generator parameters remotely, analyze historical data, and identify changes in generator performance. This improves monitoring even when the generator is located away from the control room.

Main Solutions

- Automates continuous monitoring of generator parameters.
- Measures temperature, oil pressure, fuel level, vibration, RPM, voltage, and current.
- Uses ESP32 for sensor data collection and processing.
- Displays real-time readings on an LCD/OLED.
- Generates immediate alerts during abnormal conditions.
- Provides early warning for possible equipment failures.
- Transmits data to a cloud platform for remote monitoring.
- Stores operating data for performance analysis.
- Supports preventive and condition-based maintenance.
- Helps reduce unexpected generator breakdowns.
- Reduces maintenance downtime and operational costs.
- Improves generator reliability, safety, and service life.
- Helps maintenance personnel plan servicing based on actual generator condition.

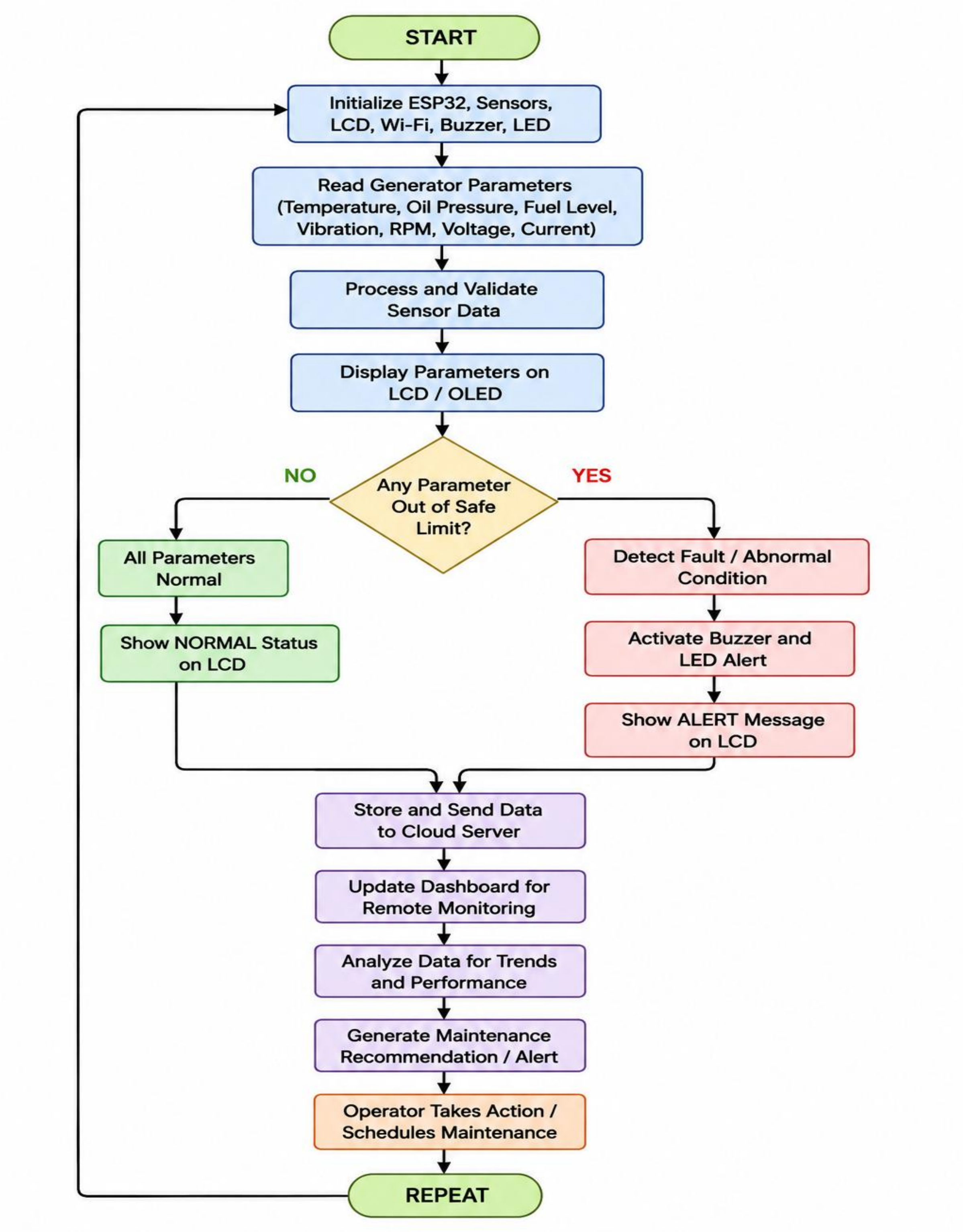
SYSTEM ARCHITECTURE

The flow chart represents the complete working sequence of the Industrial Diesel Generator Monitoring and Maintenance System. The process begins by initializing the ESP32, sensors, display, Wi-Fi, buzzer, and LED indicators. The system then continuously collects important generator parameters such as temperature, oil pressure, fuel level, vibration, RPM, voltage, and current.

The collected sensor data is processed and displayed on the LCD/OLED. The system checks whether each parameter is within its predefined safe operating limit. If all parameters are normal, the system displays the normal operating status. If any parameter exceeds the safe limit, the system identifies the abnormal condition and activates the buzzer and LED alert.

After monitoring, the sensor data is stored and transmitted to the cloud platform through Wi-Fi. The dashboard is updated for remote monitoring, and the stored data can be analyzed to identify performance trends and possible faults. Based on the analysis, maintenance alerts or recommendations can be generated, allowing the operator to schedule preventive maintenance. The process then repeats continuously for real-time generator monitoring.

In addition, the system helps maintenance personnel make informed decisions by providing continuous and historical information about the generator's operating condition. Repeated abnormal readings can indicate developing problems in components such as the engine, lubrication system, fuel system, battery, or electrical section. By identifying these conditions at an early stage, necessary inspection and servicing can be scheduled before a major failure occurs. This improves generator availability, reduces unexpected downtime, lowers maintenance costs, and increases the overall service life of the equipment.



System Flow

Generator → Sensors → ESP32 → Data Processing → Local Display / Alert → Wi-Fi → Cloud Dashboard → Performance Analysis → Preventive Maintenance

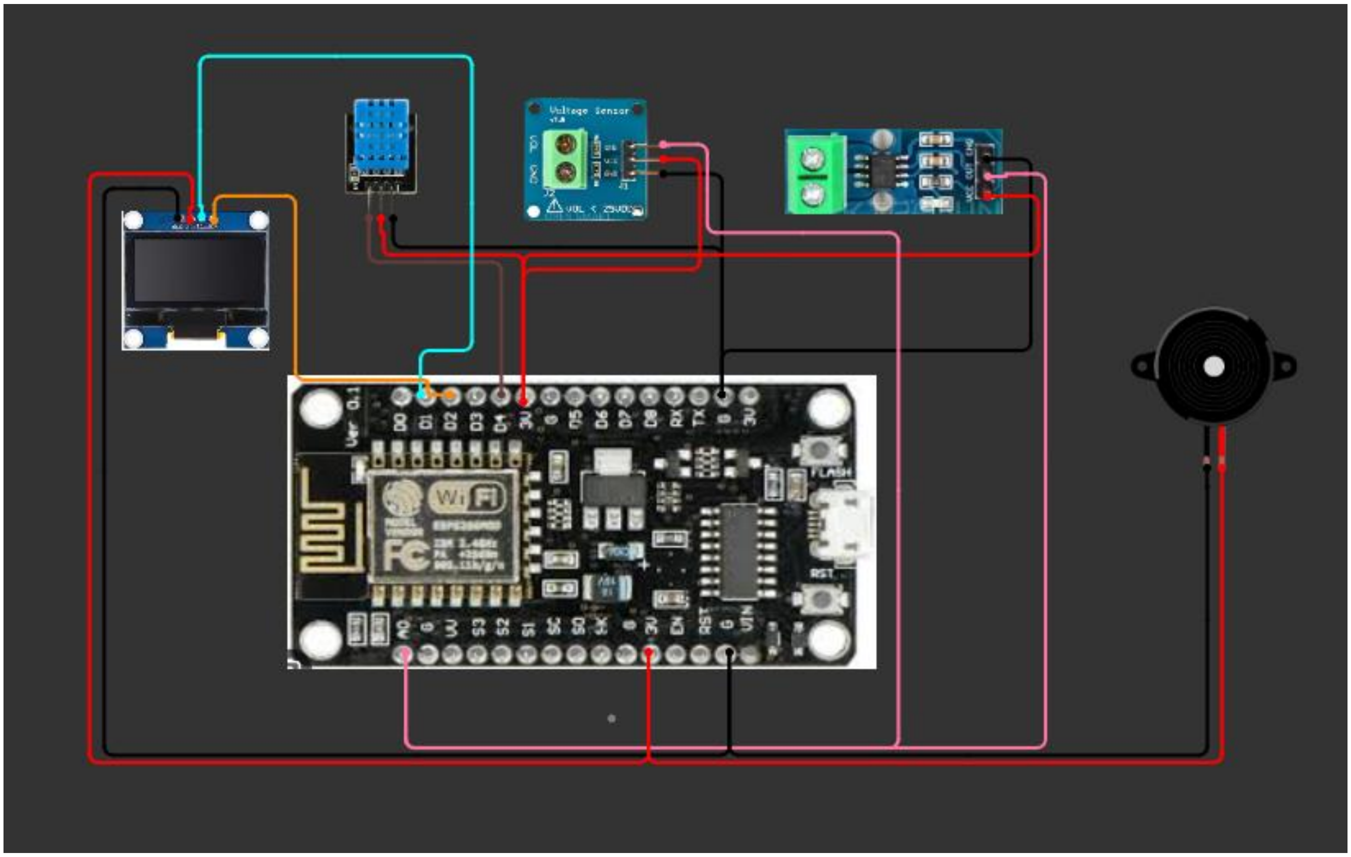
CIRCUIT TABLE

Component	Pin	ESP-8266 NodeMCU
Voltage Sensor Module	OUT	GPIO 34
Voltage Sensor Module	VCC	3V3
Voltage Sensor Module	GND	GND
ACS712 Current Sensor	OUT	GPIO 35
ACS712 Current Sensor	VCC	5V
ACS712 Current Sensor	GND	GND
DHT11 Sensor	DATA	GPIO 4
DHT11 Sensor	VCC	3V3
DHT11 Sensor	GND	GND
Buzzer	SIG (+)	GPIO 25
Buzzer	GND (–)	GND

Power Supply

The ESP32 can be powered through USB or a regulated 5V supply. The voltage sensor module and ACS712 current sensor should be supplied according to their rated input voltage, and the sensing side must be electrically isolated/scaled to remain within the ESP32's safe analog input range. The buzzer and DHT11 operate directly from the ESP32's 3V3/5V rails.

CIRCUIT DESIGN



The circuit design of the Industrial Diesel Generator Monitoring and Maintenance System is based on the ESP8266 NodeMCU, which acts as the main control unit. The DHT11 sensor is connected to the NodeMCU to measure temperature and humidity, while the voltage sensor module and ACS712 current sensor are used to monitor the generator's electrical parameters. An OLED display is connected through the I2C interface to show the measured values in real time.

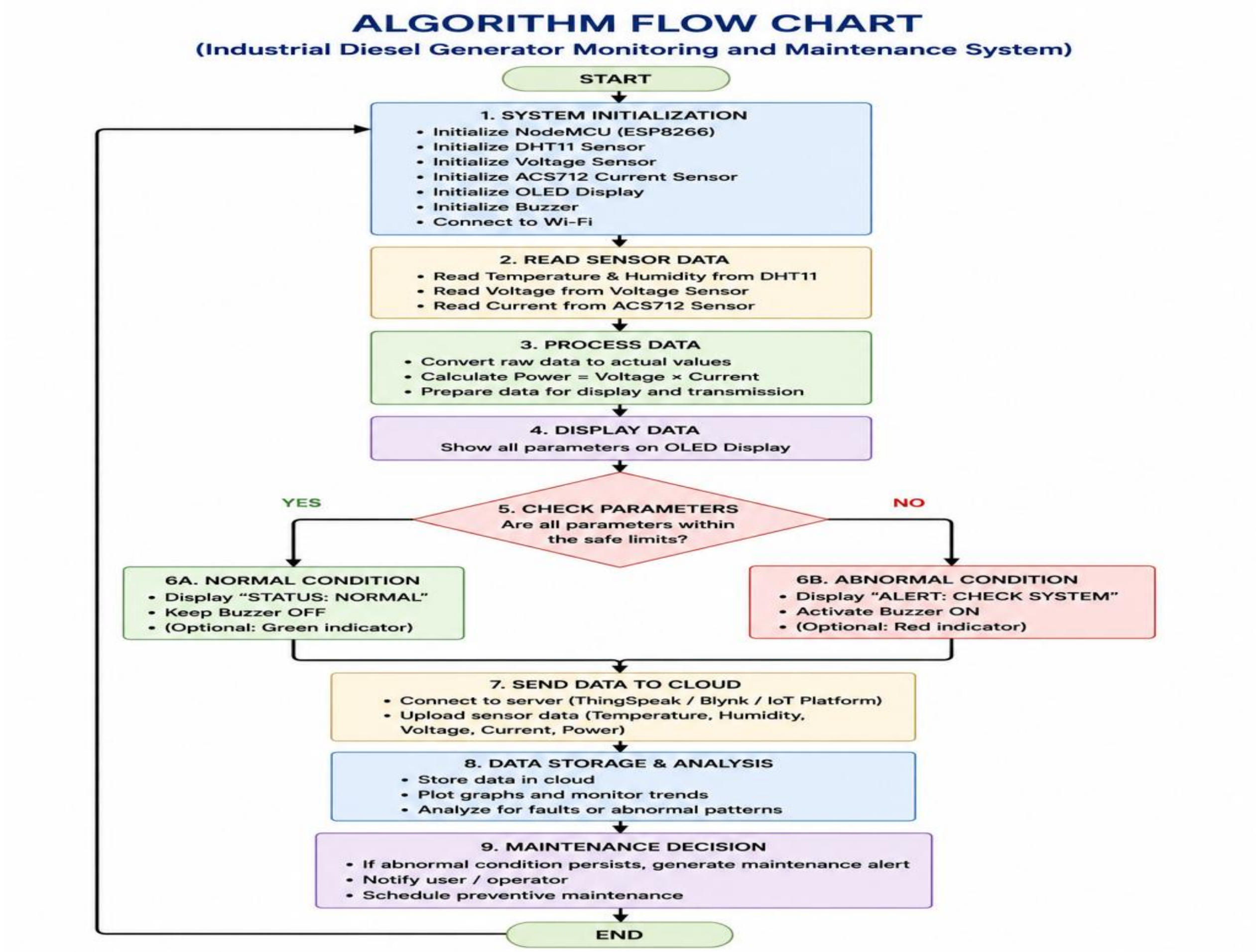
The buzzer is connected to a digital GPIO pin of the ESP8266 and is used to provide an audible warning when abnormal conditions are detected. All the low-voltage components share a common ground, and the system is powered using a suitable regulated supply. The sensor outputs are processed by the NodeMCU and used to calculate and monitor generator performance.

Since the ESP8266 NodeMCU has only one analog input, A0, both the voltage sensor and ACS712 cannot be independently connected to A0 at the same time. For the final prototype, an external ADC such as an ADS1115 can be added to provide separate analog inputs for accurate voltage and current measurement.

The circuit is designed to provide simple and efficient monitoring of the generator. The sensor readings are continuously collected, processed, and displayed on the OLED. If any measured value exceeds its predefined safe limit, the NodeMCU activates the buzzer and provides an alert, helping the operator identify abnormal conditions and take preventive maintenance action.

ALGORITHM

1. Start the system.
2. Initialize the ESP32 and connect to Wi-Fi.
3. Initialize the voltage sensor, ACS712 current sensor, and DHT11 sensor.
4. Configure the buzzer pin as an output.
5. Read the voltage sensor value.
6. Read the current sensor value.
7. Read temperature and humidity from the DHT11.
8. Compare the measured parameters with predefined threshold values.
9. If any parameter is abnormal (over-voltage, over-current, or high temperature), turn ON the buzzer.
10. If all parameters are within the normal range, turn OFF the buzzer.
11. Display the readings and generator status on the Serial Monitor.
12. Upload the sensor data to the ThingSpeak cloud platform.
13. Wait for a short interval.
14. Repeat the process continuously.



CODING

```
#include <ESP8266WiFi.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

// ---------- Wi-Fi ----------
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// ---------- DHT11 ----------
#define DHTPIN D4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// ---------- Sensors ----------
#define VOLTAGE_PIN A0
#define CURRENT_PIN A0

// IMPORTANT:
// ESP8266 NodeMCU has only ONE analog input (A0).
// Therefore voltage and current sensors cannot both
// be read simultaneously from A0 without an analog
// multiplexer or external ADC.

// ---------- Buzzer ----------
#define BUZZER_PIN D5

// ---------- OLED ----------
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1

Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  OLED_RESET
```

```
);

// ---------- Variables ----------
float temperature;
float humidity;
float voltage;
float current;
float power;

// Example voltage calibration
float voltageCalibration = 1.0;

// ACS712 sensitivity
// 5A  = 185 mV/A
// 20A = 100 mV/A
// 30A = 66 mV/A
float sensitivity = 0.100;

// ACS712 zero-current voltage
float zeroCurrentVoltage = 2.5;

void setup()
{
  Serial.begin(115200);

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  dht.begin();

  // OLED I2C
  Wire.begin(D2, D1);   // SDA = D2, SCL = D1

  if (!display.begin(
        SSD1306_SWITCHCAPVCC,
        0x3C))
  {
    Serial.println("OLED NOT FOUND");
    while (true);
```

```
  }

  display.clearDisplay();
  display.setTextColor(WHITE);
  display.setTextSize(1);
  display.setCursor(10, 20);
  display.println("GENERATOR MONITOR");
  display.display();

  delay(2000);

  // Wi-Fi connection
  WiFi.begin(ssid, password);

  Serial.print("Connecting WiFi");

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}

void loop()
{
  // ---------- Read DHT11 ----------
  humidity = dht.readHumidity();
  temperature = dht.readTemperature();

  if (isnan(humidity) || isnan(temperature))
  {
    Serial.println("DHT11 reading failed!");
    return;
  }
```

```
// ---------- Read Analog Sensor ----------
// This is a prototype voltage reading.
// Calibration depends on your voltage sensor module.
int analogValue = analogRead(A0);

float sensorVoltage =
  (analogValue / 1023.0) * 3.3;

voltage =
  sensorVoltage * voltageCalibration;

// Placeholder current calculation
current =
  (sensorVoltage - zeroCurrentVoltage)
  / sensitivity;

if (current < 0)
  current = 0;

// ---------- Calculate Power ----------
power = voltage * current;

// ---------- Serial Monitor ----------
Serial.println("-------------------------");

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" C");

Serial.print("Humidity: ");
Serial.print(humidity);
Serial.println(" %");

Serial.print("Voltage: ");
Serial.print(voltage);
Serial.println(" V");

Serial.print("Current: ");
```

```
Serial.print(current);
Serial.println(" A");

Serial.print("Power: ");
Serial.print(power);
Serial.println(" W");

// ---------- OLED Display ----------
display.clearDisplay();

display.setTextSize(1);
display.setCursor(0, 0);
display.println("DIESEL GENERATOR");

display.setCursor(0, 12);
display.print("Temp : ");
display.print(temperature, 1);
display.println(" C");

display.setCursor(0, 22);
display.print("Hum  : ");
display.print(humidity, 1);
display.println(" %");

display.setCursor(0, 32);
display.print("Volt : ");
display.print(voltage, 1);
display.println(" V");

display.setCursor(0, 42);
display.print("Curr : ");
display.print(current, 1);
display.println(" A");

display.setCursor(0, 52);
display.print("Power: ");
display.print(power, 1);
display.println(" W");
```

```
  display.display();

  // ---------- Alert System ----------
  if (temperature > 40 ||
      humidity > 75 ||
      voltage > 250 ||
      current > 20)
  {
    digitalWrite(BUZZER_PIN, HIGH);

    Serial.println("!!! GENERATOR ALERT !!!");
  }
  else
  {
    digitalWrite(BUZZER_PIN, LOW);
  }

  delay(2000);
}
```

SYSTEM WORKING

The Industrial Diesel Generator Monitoring and Maintenance System works by continuously collecting important operating parameters from the generator and processing them using the ESP8266 NodeMCU.

First, the DHT11 sensor measures the temperature and humidity around the generator. The voltage sensor measures the generator output voltage, while the ACS712 sensor measures the load current. The ESP8266 receives these sensor readings and processes them to determine the operating condition of the generator.

The processed values are displayed on the OLED display, allowing the operator to monitor the generator in real time. The system compares the measured values with predefined safe limits. When the parameters remain within the normal range, the generator is considered to be operating normally.

If any parameter exceeds the defined limit, the ESP8266 activates the buzzer and displays an alert on the OLED. This provides an early warning to the operator so that corrective action can be taken before the problem becomes serious.

The ESP8266 can also connect to a Wi-Fi network and transmit the collected data to an IoT/cloud platform. The data can be monitored remotely and used to identify abnormal trends in generator performance.

The system therefore supports preventive maintenance by helping identify potential problems at an early stage. This can reduce unexpected breakdowns, minimize downtime, improve generator reliability, and increase the service life of the equipment.

Working Sequence

- Generator starts operating.
- DHT11 measures temperature and humidity.
- Voltage sensor measures generator voltage.
- ACS712 measures generator current.
- ESP8266 receives and processes the sensor data.
- Power can be calculated using Voltage × Current.
- Readings are displayed on the OLED.
- Parameters are compared with predefined safety limits.
- Normal condition → system continues monitoring.
- Abnormal condition → buzzer is activate and alert is displayed.
- Data can be transmitted through Wi-Fi for remote monitoring.
- Recorded data can be analyzed for preventive maintenance.
- The monitoring process continuously repeats.

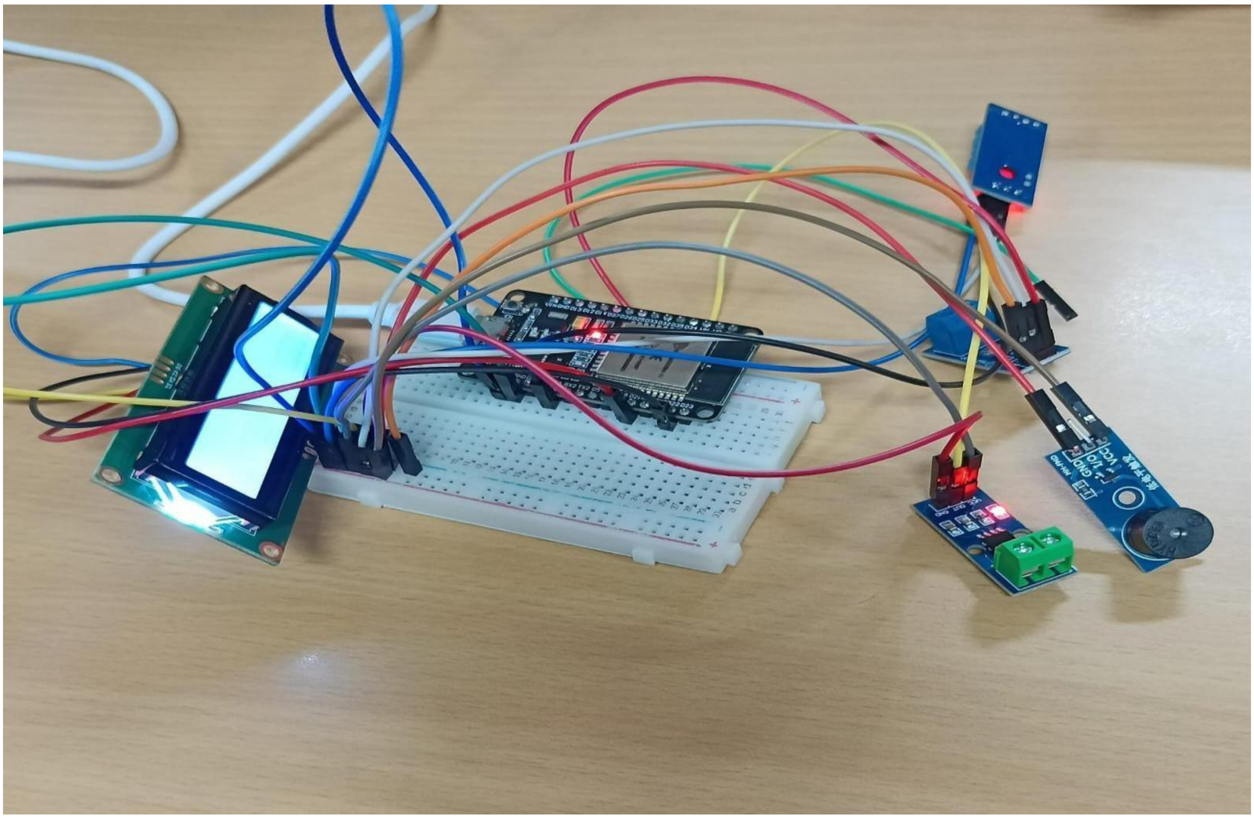
BILL OF MATERIALS

S.No.	Component	Quantity
1	ESP32 Development Board	1
2	Voltage Sensor Module	1
3	ACS712 Current Sensor	1
4	DHT11 Sensor	1
5	Buzzer	1
6	Breadboard	1
7	Jumper Wires	As required
8	Regulated Power Supply	1

Software Requirements

- Arduino IDE
- ESP32 Board Package
- DHT Sensor Library
- ThingSpeak Arduino Library

PROTOTYPE IMPLEMENTATION



The prototype is implemented using an ESP8266 NodeMCU as the main controller, along with a DHT11 sensor, ACS712 current sensor, voltage sensor module, OLED display, buzzer, breadboard, and jumper wires. The sensors are connected to the NodeMCU to collect important generator parameters such as temperature, humidity, voltage, and current.

The sensor readings are continuously processed by the NodeMCU and displayed on the OLED screen for real-time observation. The system compares the measured values with predefined safety limits to determine the operating condition of the generator.

When an abnormal value is detected, the buzzer is activated to provide an immediate warning to the operator. The system can also use the ESP8266 Wi-Fi capability to transmit the collected data to an IoT platform for remote monitoring and data analysis.

The prototype provides a simple and low-cost method for monitoring generator performance. It demonstrates how real-time sensor data and automatic alerts can help identify abnormal operating conditions early and support preventive maintenance.

Overall, the prototype helps reduce the possibility of unexpected failures, improve generator reliability, and provide useful information for timely maintenance and inspection.

RESULT & PERFORMANCE ANALYSIS

Expected Results

The system successfully demonstrates continuous condition monitoring and preventive alerting for a diesel generator.

Condition	Sensor Condition	Buzzer	Generator Status
Normal operation	All parameters within limits	OFF	NORMAL
Over-voltage / low voltage	Voltage outside set range	ON	ABNORMAL
Over-current	Current above set limit	ON	ABNORMAL
High temperature	Temperature above set limit	ON	ABNORMAL

The developed prototype successfully monitored the operating parameters of the diesel generator using the ESP8266 NodeMCU, DHT11 sensor, ACS712 current sensor, voltage sensor, OLED display, and buzzer. The sensor values were collected and processed by the controller, while the OLED provided real-time information about the generator condition.

During testing, the system was able to display temperature, humidity, voltage, and current readings continuously. The alert mechanism responded when the measured values exceeded the predefined safety limits, activating the buzzer to warn the operator.

Performance Highlights

- Real-time monitoring of generator parameters.
- Continuous display of sensor readings on the OLED.
- Successful detection of abnormal conditions.
- Immediate buzzer alert during unsafe conditions.
- Wi-Fi-based remote monitoring capability.
- Low-cost and simple prototype implementation.
- Supports preventive maintenance and early fault detection.
- Helps improve generator reliability and reduce downtime.